TRANSISTOR

Q1. A transistor has ……

1. one pn junction
2. **two pn junctions**
3. three pn junctions
4. four pn junctions

Q2. The number of depletion layers in a transistor is ……

1. four
2. three
3. one
4. **two**

Q3. The base of a transistor is ……. doped

1. heavily
2. moderately
3. **lightly**
4. none of the above

Q4. The element that has the biggest size in a transistor is ……..

1. **collector**
2. base
3. emitter
4. collector-base-junction

Q5. In a pnp transistor, the current carriers are …….

1. acceptor ions
2. donor ions
3. free electrons
4. **holes**

Q6. The collector of a transistor is ……. doped

1. heavily
2. **moderately**
3. lightly
4. none of the above

Q7. A transistor is a …… operated device

1. **current**
2. voltage
3. both voltage and current
4. none of the above

Q8. In a npn transistor, ……. are the minority carriers

1. free electrons
2. **holes**
3. donor ions
4. acceptor ions

Q9. The emitter of a transistor is …… doped

1. lightly
2. **heavily**
3. moderately
4. none of the above

Q10. In a transistor, the base current is about …….. of emitter current

1. 25%
2. 20%
3. 35 %
4. **5%**

Q11. At the base-emitter junctions of a transistor, one finds ……

1. a reverse bias
2. a wide depletion layer
3. **low resistance**
4. none of the above

Q12. The input impedance of a transistor is …….

1. high
2. **low**
3. very high
4. almost zero

Q13. Most of the majority carriers from the emitter …….

1. recombine in the base
2. recombine in the emitter
3. **pass through the base region to the collector**
4. none of the above

Q14. The current I_B is ……

1. **electron current**
2. hole current
3. donor ion current
4. acceptor ion current

Q15. In a transistor …….

$I_C = I_E + I_B$

$I_B = I_C + I_E$

$I_E = I_C - I_B$

$I_E = I_C + I_B$

Q16. The value of α of a transistor is …….

- more than 1
- **less than 1**
- 1
- none of the above

Q17. $I_C = \alpha I_E +$ ……

1. I_B
2. I_{CEO}
3. **I_{CBO}**
4. βI_B

Q18. The output impedance of a transistor is ……..

1. **high**
2. zero
3. low
4. very low

Q19. In a tansistor, I_C = 100 mA and I_E = 100.2 mA. The value of β is ……

1. 100
2. 50
3. about 1
4. **200**

Q20. In a transistor if β = 100 and collector current is 10 mA, then I_E is ……

1. 100 mA
2. **100.1 mA**
3. 110 mA
4. none of the above

Q21. The relation between β and α is ……..

1. $\beta = 1 / (1 - \alpha)$
2. **$\beta = (1 - \alpha) / \alpha$**
3. **$\beta = \alpha / (1 - \alpha)$**
4. $\beta = \alpha / (1 + \alpha)$

Q22. The value of β for a transistor is generally ……..

1. 1
2. less than 1

3. **between 20 and 500**
4. above 500

Q23. The most commonly used transistor arrangement is …… arrangement

1. **common emitter**
2. common base
3. common collector
4. none of the above

Q24. The input impedance of a transistor connected in …….. arrangement is the highest

1. common emitter
2. **common collector**
3. common base
4. none of the above

Q25. The output impedance of a transistor connected in ……. arrangement is the highest

1. common emitter
2. common collector
3. **common base**
4. none of the above

Q26. The phase difference between the input and output voltages in a common base arrangement is …….

1. 180°
2. 90°
3. 270°
4. **0°**

Q27. The power gain in a transistor connected in ……. arrangement is the highest

1. **common emitter**
2. common base
3. common collector
4. none of the above

Q28. The phase difference between the input and output voltages of a transistor connected in common emitter arrangement is ……

1. **0°**
2. **180°**
3. 90°
4. 270°

Q29. The voltage gain in a transistor connected in ……. arrangement is the highest

1. common base
2. **common collector**
3. **common emitter**
4. none of the above

Q30. As the temperature of a transistor goes up, the base-emitter resistance ……

1. **decreases**
2. increases
3. remains the same
4. none of the above

Answer : 1

Q31. The voltage gain of a transistor connected in common collector arrangement is …….

1. equal to 1
2. more than 10
3. more than 100
4. **less than 1**

Q32. The phase difference between the input and output voltages of a transistor connected in common collector arrangement is ……

1. 180°
2. **0°**
3. 90°
4. 270°

Q33. $I_C = \beta I_B +$ …….

1. I_{CBO}
2. I_C
3. **I_{CEO}**
4. αI_E

Q34. $I_C = [\alpha / (1 - \alpha)] I_B +$ ……

1. **I_{CEO}**
2. I_{CBO}
3. I_C
4. $(1 - \alpha) I_B$

Q35. $I_C = [\alpha / (1 - \alpha)] I_B + [\ldots\ldots / (1 - \alpha)]$

1. **I_{CBO}**
2. I_{CEO}
3. I_C
4. I_E

Q36. BC 147 transistor indicates that it is made of ……..

1. germanium
2. **silicon**
3. carbon
4. none of the above

Q37. I_{CEO} = (……) I_{CBO}

1. β
2. 1 + α
3. **1 + β**
4. none of the above

Q38. A transistor is connected in CB mode. If it is not connected in CE mode with same bias voltages, the values of I_E, I_B and I_C will ……..

1. **remain the same**
2. increase
3. decrease
4. none of the above

Q39. If the value of α is 0.9, then value of β is ……..

1. 9
2. 0.9
3. 900
4. **90**

Q40. In a transistor, signal is transferred from a …… circuit

1. high resistance to low resistance
2. **low resistance to high resistance**
3. high resistance to high resistance
4. low resistance to low resistance

Q41. The arrow in the symbol of a transistor indicates the direction of …….

1. electron current in the emitter
2. electron current in the collector
3. **hole current in the emitter**
4. donor ion current

Q42. The leakage current in CE arrangement is ……. that in CB arrangement

1. **more than**
2. less than
3. the same as
4. none of the above

Q43. A heat sink is generally used with a transistor to ……

1. increase the forward current
2. decrease the forward current
3. compensate for excessive doping
4. **prevent excessive temperature rise**

Q44. The most commonly used semiconductor in the manufacture of a transistor is …….

1. germanium
2. **silicon**
3. carbon
4. none of the above

Q45. The collector-base junction in a transistor has ……..

1. forward bias at all times
2. **reverse bias at all times**
3. low resistance
4. none of the above

Q46. When transistors are used in digital circuits they usually operate in the …….

1. active region
2. breakdown region
3. **saturation and cutoff regions**
4. linear region

Q47. Three different Q points are shown on a dc load line. The upper Q point represents the …….

1. minimum current gain
2. intermediate current gain
3. maximum current gain
4. cutoff point

Q48. A transistor has a β_{DC} of 250 and a base current, I_B, of 20 μ A. The collector current, I_C, equals to …….

1. 500 μA

2. **5 mA**
3. 50 mA
4. 5 A

Q49. A current ratio of I_C/I_E is usually less than one and is called ……

1. beta
2. theta
3. **alpha**
4. omega

Q50. With the positive probe on an NPN base, an ohmmeter reading between the other transistor terminals should be ……

1. open
2. infinite
3. **low resistance**
4. high resistance

Q51. In a CE configuration, an emitter resistor is used for ……

1. **stabilization**
2. ac signal bypass
3. collector bias
4. higher gain

Q52. Voltage-divider bias provides …….

1. an unstable Q point
2. **a stable Q point**
3. a Q point that easily varies with changes in the transistor's current gain
4. a Q point that is stable and easily varies with changes in the transistor's current gain

Q53. To operate properly, a transistor's base-emitter junction must be forward biased with reverse bias applied to which junction?

1. collector-emitter
2. base-collector
3. base-emitter
4. **collector-base**

Q54. The ends of a load line drawn on a family of curves determine ……

1. **saturation and cutoff**
2. the operating point
3. the power curve
4. the amplification factor

Q55. If V_{CC} = +18 V, voltage-divider resistor R_1 is 4.7 kΩ, and R_2 is 1500 Ω, then the base bias voltage is …….

1. 8.7 V
2. **4.35 V**
3. 2.9 V
4. 0.7 V

Q56. The C-B configuration is used to provide which type of gain?

1. **voltage**
2. current
3. resistance
4. power

Q57. The Q point on a load line may be used to determine ……

1. V_C
2. V_{CC}
3. **V_B**
4. I_C

Q58. A transistor may be used as a switching device or as a ……

1. fixed resistor
2. tuning device
3. rectifier
4. **variable resistor**

Q59. If an input signal ranges from 20–40 μA (microamps), with an output signal ranging from .5–1.5 mA (milliamps), what is the ac beta?

1. 0.05
2. 20
3. **50**
4. 500

Q60. Beta's current ratio is ……..

1. **I_C/I_B**
2. I_C/I_E
3. I_B/I_E
4. I_E/I_B

Q61. A collector characteristic curve is a graph showing ……..

1. emitter current (I_E) versus collector-emitter voltage (V_{CE}) with (V_{BB}) base bias voltage held constant
2. **collector current (I_C) versus collector-emitter voltage (V_{CE}) with (V_{BB}) base bias voltage held constant**

3. collector current (I_C) versus collector-emitter voltage (V_C) with (V_{BB}) base bias voltage held constant
4. collector current (I_C) versus collector-emitter voltage (V_{CC}) with (V_{BB}) base bias voltage held constant

Q62. With low-power transistor packages, the base terminal is usually the …….

1. tab end
2. **middle**
3. right end
4. stud mount

Q63. When a silicon diode is forward biased, V_{BE} for a CE configuration is ……..

1. voltage-divider bias
2. 0.4 V
3. **0.7 V**
4. emitter voltage

Q64. What is the current gain for a common-base configuration where I_E = 4.2 mA and I_C = 4.0 mA?

1. 16.8
2. 1.05
3. 0.2
4. **0.95**

Q65. With a PNP circuit, the most positive voltage is probably ……

1. **ground**
2. V_C
3. V_{BE}
4. V_{CC}

Q66. If a 2 mV signal produces a 2 V output, what is the voltage gain?

1. 0.001
2. 0.004
3. **100**
4. **1000**

Q67. Most of the electrons in the base of an NPN transistor flow ……

1. out of the base lead
2. **into the collector**
3. into the emitter
4. into the base supply

Q68. In a transistor, collector current is controlled by ……..

1. collector voltage
2. **base current**
3. collector resistance
4. all of the above

Q69. Total emitter current is ……

1. $I_E - I_C$
2. $I_C + I_E$
3. **$I_B + I_C$**
4. $I_B - I_C$

Q70. Often a common-collector will be the last stage before the load; the main function(s) of this stage is to …….

1. provide voltage gain
2. provide phase inversion
3. provide a high-frequency path to improve the frequency response
4. **buffer the voltage amplifiers from the low-resistance load and provide impedance matching for maximum power transfer**

Q71. For a CC configuration to operate properly, the collector-base junction should be reverse biased, while forward bias should be applied to …… junction.

1. **collector-emitter**
2. base-emitter
3. collector-base
4. cathode-anode

Q72. The input/output relationship of the common-collector and common-base amplifiers is ……..

1. 270 degrees
2. 180 degrees
3. 90 degrees
4. **0 degrees**

Q73. If a transistor operates at the middle of the dc load line, a decrease in the current gain will move the Q point …….

1. off the load line

2. nowhere
3. up
4. **down**

Q74. Which is the higher gain provided by a CE configuration?

1. voltage
2. current
3. resistance
4. **power**

Q75. What is the collector current for a CE configuration with a beta of 100 and a base current of 30 μA?

1. 30 μA
2. 0.3 μA
3. **3 mA**
4. 3 MA

……..